ANDERSON JUNIOR COLLEGE

Preliminary Examination 2010

CHEMISTRY **8872/01**

Higher 1 **23 September 2010**

Paper 1 Multiple Choice **50 minutes**

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Multiple Choice Answer Sheet

Write your name, PDG and NRIC/FIN number. **DO NOT** include the reference letter.

Shade the NRIC / FIN number.

Exam Title: JC2 Prelim

Exam Details: H1 Chem / Paper 1

Date: 23/09/2010

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1 Which statement about one mole of a metal is always true?

A It contains the same number of particles as one mole of hydrogen atoms.

B It contains the same number of particles as 1/12 mole of ^{12}C.

C It has the same mass as one mole of hydrogen atoms.

D It is liberated by one mole of electrons.

2 *Use of the Data Booklet is relevant to this question.*

A garden fertiliser is said to have a phosphorus content of 30.0% by mass 'P_2O_5 soluble in water'.

What is the percentage by mass of phosphorus in the fertiliser?

A 3.93% **B** 6.55% **C** 13.1 % **D** 26.2%

3 *Use of the Data Booklet is relevant to this question.*

The ^{68}Ge isotope of the Group IV element germanium is medically useful because it undergoes a natural radioactive process to give a gallium isotope, ^{68}Ga, which can be used to detect tumors. This transformation of germanium occurs when one of its electrons enters the nucleus, changing a proton into a neutron.

What do the isotopes ^{68}Ga and ^{68}Ge have in common?

A Both isotopes have 37 neutrons in their nuclei.

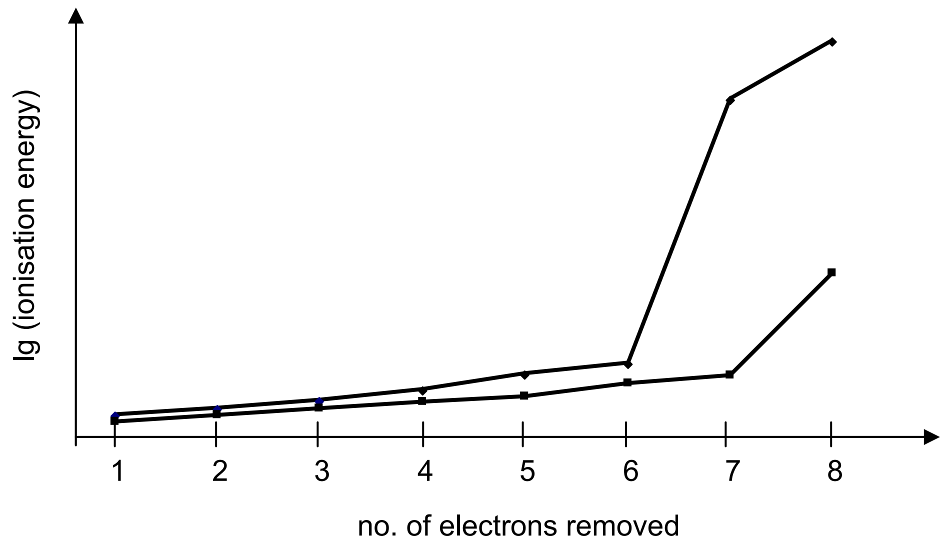
B Both isotopes have more electrons than protons.

C Both isotopes have an outer electronic configuration $4s^2\ 4p^3$.

D Both isotopes contain the same number of nucleons in their nuclei.

4 *Use of the Data Booklet is relevant to this question.*

The graph shows the logarithm, lg, of the first eight ionisation energies of two elements in Periods 2 and 3.



What is the most likely compound that will be formed between the two elements?

A OF_2 **B** OCl_2 **C** SF_2 **D** SCl_2

5 In which of the following pairs does the first species have a larger bond angle than the second?

A CH_4, CH_3^+

B NCl_3, BH_3

C XeF_4, SF_6

D H_2O, H_2S

6 Which statement concerning the lattice structure of graphite and diamond is **incorrect**?

A All C–C bonds in graphite and diamond are formed from sp^3–sp^3 overlap.

B Graphite is energetically more stable than diamond.

C The bond energy of the C–C bonds in graphite is greater than that in diamond.

D The C–C–C bond angle between nearest neighbours is smaller in diamond than in graphite.

7 In which of the following do all three compounds have giant lattice structure?

A BaO, AlF_3, SiO_2

B K_2O, BCl_3, Cr_2O_3

C MgO, Fe_2O_3, P_4O_{10}

D Na_2O, SiO_2, SO_3

8 When 25 cm^3 of 1.0 mol dm^{-3} sodium hydroxide is completely neutralised by 1.0 mol dm^{-3} hydrochloric acid, the temperature of the mixture rose by 6.8 °C. What will be the temperature change if 50 cm^3 of 0.5 mol dm^{-3} sodium hydroxide is completely neutralised by 0.5 mol dm^{-3} hydrochloric acid? (Assume heat loss to be negligible in each case)

A 1.7 °C

B 3.4 °C

C 6.8 °C

D 13.6 °C

9 Titanium occurs naturally in the mineral rutile, TiO_2. Rutile is a major ore of titanium, a metal used for high tech alloys because of its light weight, high strength and resistance to corrosion.

One possible method suggested for the extraction of the metal is to reduce the rutile by heating it with carbon:

$$TiO_2(s) + 2C(s) \rightarrow Ti(s) + 2CO(g)$$

Given that $\Delta H_f^{\ominus}[TiO_2(s)] = -940$ kJ mol^{-1} and $\Delta H_f^{\ominus}[CO(g)] = -110$ kJ mol^{-1}, what is the ΔH for this reaction?

A +720 kJ mol^{-1}

B +830 kJ mol^{-1}

C –830 kJ mol^{-1}

D It cannot be calculated from the information above.

10 The nickel-cadmium rechargeable battery is based upon the following overall reaction.

$$Cd + 2NiOOH + 4H_2O \rightarrow Cd(OH)_2 + 2Ni(OH)_2.H_2O$$

What is the oxidation number of nickel at the beginning and at the end of the reaction?

	beginning	end
A	+1.5	+2
B	+2	+3
C	+3	+2
D	+3	+4

11 In the Haber-Bosch Process, ammonia gas is synthesised by reacting nitrogen gas with hydrogen gas.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta H^\ominus = -46 \text{ kJ mol}^{-1}$$

If the temperature of an equilibrium mixture of the three gases is increased at constant pressure, will the volume of the mixture increase or decrease and why?

A The volume will increase due to only thermal expansion.

B The volume will increase due to a shift of equilibrium to the left and thermal expansion.

C The volume will remain the same as any thermal expansion could be offset by any shifting of equilibrium to the right.

D The volume will decrease as a shift of equilibrium to the right could more than offset any thermal expansion.

12 When 0.1 mol of of bismuth chloride is added to 2 dm^3 of water, it reacts to form 0.02 mol of white precipitate of bismuth oxychloride and a solution of hydrochloric acid.

The equation for the reaction is as follows.

$$BiCl_3(aq) + H_2O(l) \rightleftharpoons BiOCl(s) + 2HCl(aq)$$

What is the correct expression for the equilibrium constant K_c?

A $\dfrac{(2 \times 0.02)^2}{0.08}$

B $\dfrac{(0.02)(2 \times 0.02)^2}{0.08}$

C $\dfrac{\left(\dfrac{2 \times 0.02}{2}\right)^2}{\dfrac{0.08}{2}}$

D $\dfrac{\left(\dfrac{0.02}{2}\right)\left(\dfrac{2 \times 0.02}{2}\right)^2}{\dfrac{0.08}{2}}$

13 What is the final pH of the solution formed by mixing equal volumes of two separate solutions of a strong acid with pH 2.0 and pH 3.0?

A 1.96 **B** 2.26 **C** 2.50 **D** 2.74

14 The reaction of acidified, aqueous potassium iodide with aqueous hydrogen peroxide is shown below.

$$2I^-(aq) + H_2O_2(aq) + 2H^+(aq) \rightarrow I_2(aq) + 2H_2O(l)$$

The rate equation is rate = $k[H_2O_2][I^-]$. Which of the following conclusion **cannot** be drawn from this information?

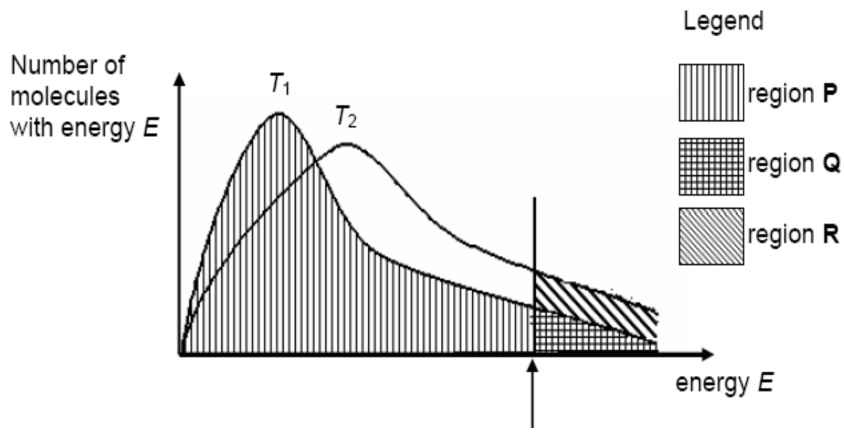
A The iodide ion is oxidised by the hydrogen peroxide.

B The colour of the solution turns brownish at the end of reaction.

C The overall order of reaction is two.

D The acid acts as catalyst.

15 The distribution of the number of molecules with energy E is given in the sketch for two temperatures, T_1 and a higher temperature T_2. The letters, P, Q, and R refer to the separate and differently shaded areas. The activation energy is marked on the energy axis.



Which expression gives the fraction of the molecules present which have at least the activation energy at the higher temperature T_2?

A $\dfrac{Q}{P}$ **B** $\dfrac{Q+R}{P}$ **C** $\dfrac{Q+R}{P+Q-R}$ **D** $\dfrac{Q+R}{P+Q}$

16 Consecutive elements **E**, **F** and **G** are in Period 3 of the Periodic Table. Element **F** has the lowest first ionisation energy and the highest melting point.

What could be the identities of **E**, **F** and **G**?

A magnesium, aluminium, silicon

B aluminium, silicon, phosphorus

C silicon, phosphorus, sulfur

D phosphorus, sulfur, chlorine

17 The oxide and chloride of an element **H** are separately mixed with water. The two resulting solutions have the same effect on litmus.

What is element **H**?

A sodium

B magnesium

C aluminium

D phosphorus

18 Which molecules have isomers that exhibit *cis–trans* isomerism?

I	II	III	IV
C_3H_6BrI	C_3H_5I	$C_3H_4I_2$	C_3H_4BrI

A I, II and III only

B II and IV only

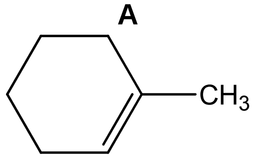
C III and IV only

D II, III and IV only

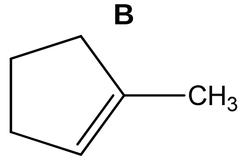
19 A hydrocarbon, on heating with an excess of hot concentrated $KMnO_4$, produces $HO_2CCH_2CH_2CH_2COCH_3$.

Which of the following **cannot** be the hydrocarbon?

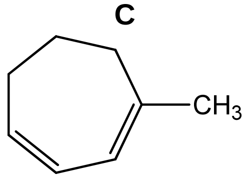
A



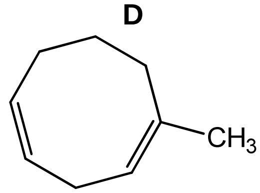
B



C



D



20 Four drops of 1–chlorobutane, 1–bromobutane, 1–iodobutane were put separately into three test–tubes containing 1.0 cm^3 of aqueous silver nitrate at 60 °C.

A hydrolysis reaction occurred. (R represents the butane chain C_4H_9– and *X* the halogen atom.

$$H_2O(l) + R–X(l) + Ag^+(aq) \rightarrow R–OH(aq) + AgX(s) + H^+(aq)$$

The rate of formation of precipitate in the tubes was in the order RC*l* < RBr < RI.

What helps to explain this observation?

A The bond energy of R–*X* decreases from RC*l* to RI.

B The ionisation energy of the halogen decreases from C*l* to I.

C The R–*X* bond polarity decreases from RC*l* to RI.

D The lattice energy of Ag*X*(s) decreases from AgC*l* to AgI.

21 Which reagent could detect the presence of added alcohol in petrol consisting mainly of a mixture of alkanes and alkenes?

A Br_2 in CCl_4

B anhydrous $SOCl_2$

C aqueous $KMnO_4$

D aqueous NaOH

22 The compound **J**, CH_3COCH_2CHO, was reacted in separate experiments with:

(i) alkaline aqueous iodine;
(ii) 2,4-dinitrophenylhydrazine;
(iii) Tollens' reagent $[Ag(NH_3)_2]^+(aq)$;
(iv) lithium aluminium hydride in dry ether.

One mole of **J** could

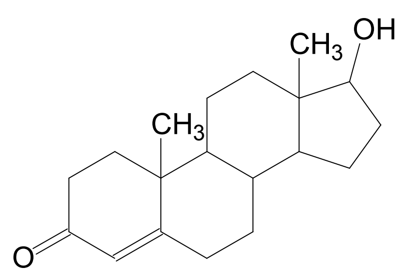
A form two moles of CHI_3 in (i).

B react with two moles of 2,4-dinitrophenylhydrazine in (ii).

C form one mole of $CH_3COCH_2CH_2OH$ and two moles of Ag in (iii).

D form a product in (iv) which further reacts with Na to produce two moles of H_2 gas.

23 Testosterone is a steroid hormone found in mammals, reptiles, birds and other vertebrates.



testosterone

Testosterone is first reacted with hydrogen in the presence of a platinum catalyst, and the product is then warmed with acidified $KMnO_4$.

Given that no carbon-carbon σ-bond is broken in this process, how many double bonds will there be in the structure of the final product?

A 0 **B** 1 **C** 2 **D** 3

24 The ester, $CH_3CH_2CH(CH_3)CH_2OCOCH_3$, has an odour of bananas.

Which set of reagents could be used to prepare this ester in the laboratory?

A $CH_3CH_2CH(CH_3)CH_2CO_2H$, CH_3Cl, H_2SO_4

B $CH_3CH_2CH(CH_3)CH_2Cl$, CH_3CO_2H, H_2SO_4

C $CH_3CH_2CH(CH_3)CH_2CO_2H$, CH_3OH, H_2SO_4

D $CH_3CH_2CH(CH_3)CH_2OH$, CH_3CO_2H , H_2SO_4

25 Which pairs of reactions could have the same common intermediate?

K	$CH_3CH_2CH_3$	$\rightarrow$	intermediate	$\rightarrow$	$(CH_3)_2CHCN$
L	$C_6H_5CO_2CH(CH_3)_2$	$\rightarrow$	intermediate	$\rightarrow$	$(CH_3)_2CHBr$
M	$CH_3CH=CH_2$	$\rightarrow$	intermediate	$\rightarrow$	$(CH_3)_2CH(NH_2)$
N	$CH_3CH_2CH_2Cl$	$\rightarrow$	intermediate	$\rightarrow$	CH_3COOH

A **K** and **L**

B **L** and **N**

C **K** and **M**

D **M** and **N**

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct.)

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26 Which of the following aqueous solutions do not considerably change in pH when relatively small volumes of strong acid or strong alkali are added?

1 A mixture of HF and NaF

2 A mixture of HCN and NaCN

3 A mixture of H_2CO_3 and $NaHCO_3$

27 Hydrazine, N_2H_4, is widely used as a rocket fuel because it reacts with oxygen as shown, producing 'environmentally friendly' gases.

$$N_2H_4(l) + O_2(g) \rightarrow N_2(g) + 2H_2O(g) \qquad \Delta H = -534\ \text{kJ mol}^{-1}$$

Despite its use as a rocket fuel, hydrazine does not spontaneously burn in oxygen.

Why does hydrazine not burn spontaneously?

1 The activation energy is too high.

2 The N≡N bond is very strong.

3 Hydrazine is a liquid.

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

28 Which of the following statements are correct for the sequence of compounds below considered from left to right?

NaF MgO A*l*N SiC

1 The formula-units of these compounds are isoelectronic (have the same number of electrons).

2 The bonding becomes increasingly covalent.

3 The electronegativity difference between the elements in each compound increases.

29 In which way are propane and propene similar?

1 They are both neutral to an indicator solution.

2 They can both produce propan-2-ol under suitable conditions.

3 They both have the same standard enthalpy change of combustion.

30 The alcohol, $C_5H_{10}O$, may be oxidised by acidified aqueous potassium dichromate (VI) to a compound C_5H_8O. Which of the following could be the structural formula of the alcohol?

1 $CH_3CHCHCH_2CH_2OH$

2 $CH_2C(CH_3)CH(OH)CH_3$

3